UNIT 3 · TOPIC 3.3

Protecting Networks: Segmentation

Quiz, Answer Key

Teacher answer key, 9 questions, 1 point each. The correct answer is highlighted; the explanation and distractor notes follow.

1. Which of the following is a technique used to segment a network?

- ○ A. Installing antivirus software on all endpoint devices
- ○ B. Enabling HTTPS on an organization's public-facing web server
- **● C. Creating VLANs to separate groups of devices into distinct logical segments**
- ○ D. Requiring multi-factor authentication for all user logins

Correct answer: C

Why: VLANs (Virtual Local Area Networks) are a core network segmentation technique that logically separate groups of devices even when they share the same physical switching infrastructure. Devices in different VLANs cannot communicate directly without passing through a Layer 3 routing device. The other options are legitimate security controls but operate at different layers, endpoint protection, transport-layer encryption, and access authentication, none of which create network-level boundaries between device groups.

*Why the others are wrong: **A**, Antivirus software is an endpoint control that protects individual devices after a threat has reached them. It does not establish any boundary between network segments. **B**, HTTPS encrypts data in transit between a client and server but does not isolate network segments from each other. All devices on a flat network remain reachable regardless of whether HTTPS is in use. **D**, MFA is an identity and authentication control. It can restrict who accesses a resource but does not create network-layer separation between device groups.*

2. A network administrator configures separate VLANs for the HR department and the Engineering department on the same physical switch. What does this configuration primarily accomplish?

- ○ A. It encrypts all traffic traveling between HR and Engineering devices.
- **● B. It separates the two departments into distinct logical network segments.**
- ○ C. It prevents external attackers from reaching either department's devices.
- ○ D. It automatically applies firewall rules to traffic between the two departments.

Correct answer: B

Why: VLANs create logical network segments on shared physical infrastructure. Devices assigned to different VLANs cannot communicate directly with each other, traffic must pass through a Layer 3 device (a router or Layer 3 switch) where access controls can be enforced. This separates the HR and Engineering broadcast domains even though both departments share the same physical switch hardware.

*Why the others are wrong: **A**, VLANs do not encrypt traffic. Encryption is a separate control applied at the transport (TLS) or application layer, independent of how the network is segmented. **C**, VLANs control internal traffic flow between segments. Protection against external attackers is handled by a separate perimeter control such as a firewall or edge router. **D**, VLAN assignment by itself does not configure firewall rules. Firewall ACLs must be explicitly created on the routing device that handles inter-VLAN traffic.*

3. Which of the following best describes a security benefit of network segmentation?

- ○ A. It eliminates the need for encryption because traffic stays within isolated segments.
- ○ B. It improves network performance by reducing the number of nodes in each broadcast domain.

● **C. It limits the spread of an attack by containing compromised devices within a single segment.**

○ D. It prevents insider threats by blocking all employee access to sensitive servers.

Correct answer: C

Why: Network segmentation limits lateral movement. If an attacker compromises one device, they are contained within that device's segment and cannot directly reach devices in other segments. The attacker must traverse a controlled inter-segment routing point, where ACLs or firewall rules can detect and block the traversal attempt, before reaching resources in a different segment. Containing the blast radius of a compromise is the primary security rationale for segmentation.

*Why the others are wrong: **A**, Segmentation does not replace encryption. Data can still be intercepted by other devices within the same segment, and traffic traversing inter-VLAN routing points is also exposed without separate encryption controls. **B**, Smaller broadcast domains do improve network performance (fewer unnecessary broadcast packets), but this is an operational benefit, not a security one. Students should distinguish performance improvements from security controls. **D**, Segmentation controls the flow of traffic between network zones, it does not block all employee access to sensitive servers. Doing so would prevent legitimate access and defeat the operational purpose of the servers.*

4. A hospital network is segmented into three VLANs: one for patient monitoring devices, one for administrative workstations, and one for the billing system. An attacker successfully installs malware on an administrative workstation. Which outcome is most likely as a direct consequence of the segmentation?

● **A. The attacker cannot reach patient monitoring devices or the billing system without traversing an inter-VLAN routing control point.**

○ B. The malware will be automatically detected and removed by the VLAN switch firmware.

○ C. The malware will be confined to the workstation's hard drive with no network access at all.

○ D. The attacker will gain access to all three VLANs because malware can forge VLAN tags.

Correct answer: A

Why: VLAN segmentation prevents direct Layer 2 communication between segments. To move from the administrative VLAN to the patient monitoring or billing VLANs, traffic must be routed through a Layer 3 device where ACLs or firewall rules can inspect and block it. The segmentation does not eliminate the threat, the admin workstation is still compromised, but it forces the attacker through a chokepoint, disrupting lateral movement and buying defenders time to detect and respond.

*Why the others are wrong: **B**, Switches enforce VLAN membership based on port assignment or 802.1Q tags. They do not perform malware scanning. Malware detection is an endpoint or IDS/IPS function, separate from switching infrastructure. **C**, The compromised workstation can still communicate with other devices within its own VLAN. Segmentation limits cross-segment movement, not all network access. The malware could still attack other admin workstations, exfiltrate data to an external server, or attempt inter-VLAN traversal through the routing device. **D**, VLAN-hopping attacks (e.g., double-tagging or switch-spoofing) are real but require specific misconfigurations and do not give automatic access to all VLANs. Well-configured segmentation does not inherently allow malware to traverse all segments.*

5. A company has a flat network where 400 employee workstations, a public web server, and an internal payroll database share the same broadcast domain. The security team's primary concern is a compromised employee workstation reaching the payroll database. Which segmentation approach best addresses this goal?

○ A. Move all employees to a single VLAN but increase logging on the payroll database server.

○ B. Encrypt all traffic on the flat network using TLS to prevent unauthorized reading of payroll data.

● C. Separate the web server, employee workstations, and payroll database into distinct VLANs with inter-VLAN traffic controlled by a firewall.

○ D. Apply a uniform strong password policy across all devices so a single compromised credential cannot reach the payroll database.

Correct answer: C

Why: Placing the web server, workstations, and payroll database in separate VLANs and routing inter-VLAN traffic through a firewall creates distinct network-level boundaries. A compromised workstation in the employee VLAN cannot directly reach the payroll database VLAN, its traffic must pass through the firewall, where a rule denying workstation-to-database connections can block the attempt. The public web server is also isolated from both internal segments, limiting the attack surface if the public server is compromised.

*Why the others are wrong: **A**, Consolidating all employees into one VLAN does not create a boundary between the workstations and the payroll database, all devices remain in the same broadcast domain. Increased logging may improve detection but does not prevent lateral movement. **B**, TLS encrypts data in transit, protecting it from interception. It does not prevent a compromised device from initiating connections to the payroll database, the device would simply use TLS to send malicious requests to it. **D**, Strong passwords reduce credential-based attacks but do not address the network path between a compromised device and the payroll database. Once a workstation is compromised via malware (not credential theft), password policy provides no lateral movement barrier.*

6. Compared to a flat network, how does VLAN-based segmentation improve network security?

○ A. It enforces 802.1X authentication on all devices, preventing unauthorized users from joining the network.

○ B. It eliminates the need for a firewall because each VLAN functions as its own security perimeter.

○ C. It automatically encrypts intra-VLAN traffic to prevent data interception within a segment.

● D. It reduces the blast radius of a security incident by limiting which devices can directly communicate with each other.

Correct answer: D

Why: A flat network allows any device to attempt communication with any other device in the same broadcast domain without crossing a security boundary. VLAN segmentation divides devices into logical groups where direct cross-group communication is blocked at the Layer 2 level. An attacker or compromised device is constrained to its own segment and must traverse an ACL- or firewall-enforced routing point to reach other segments. This reduction in the blast radius of a compromise is the core security improvement over a flat architecture.

*Why the others are wrong: **A**, 802.1X is a port-based network access control protocol, a separate mechanism from VLAN segmentation, though the two can be combined. VLAN assignment alone does not enforce 802.1X authentication. **B**, VLANs create internal boundaries but do not replace firewalls. In fact, firewalls become more important in a segmented network: they are deployed at inter-VLAN routing points to inspect and control traffic crossing segment boundaries. **C**, VLANs do not encrypt traffic. Devices within the same VLAN communicate in cleartext at Layer 2. Encryption must be applied separately at the transport or application layer.*

7. A security architect is comparing two network designs for a corporate campus. Design A uses a single flat network with a hardened perimeter firewall. Design B uses VLAN-based segmentation with per-segment access controls at each inter-VLAN routing point. The architect's primary concern is minimizing the impact of a compromised employee workstation being used by a malicious insider. Which evaluation best supports choosing Design B over Design A?

○ A. Design B is preferred because segmentation eliminates insider threats entirely by preventing all employee-to-server traffic.

○ B. Design A is preferred because a properly configured perimeter firewall is sufficient to contain threats originating from inside the network.

● C. Design B is preferred because VLAN segmentation forces intra-campus lateral movement through access-controlled routing points, reducing the set of resources reachable from any single compromised device.

○ D. Design A is preferred because VLAN segmentation increases administrative complexity without providing measurable security benefit beyond a perimeter firewall.

Correct answer: C

Why: A perimeter firewall (Design A) is designed to filter traffic entering and leaving the network boundary. It provides little or no control over lateral movement within the flat internal network, once an insider has network access, they can attempt connections to any internal resource. Design B forces all cross-segment traffic through routing points where ACLs can whitelist only legitimate inter-segment flows. NIST SP 800-125B and the broader defense-in-depth model explicitly recommend internal network segmentation to reduce lateral movement risk, perimeter controls alone are insufficient against insider threats or threats that have already bypassed the perimeter.

*Why the others are wrong: **A**, No single control eliminates insider threats, this claim is categorically too strong. Segmentation raises the cost and detectability of lateral movement; it does not block all employee-to-server traffic, which would prevent legitimate operations. **B**, A perimeter firewall filters ingress and egress traffic at the network boundary but has no visibility into host-to-host traffic within a flat internal network. An insider operating on the internal LAN operates entirely behind the perimeter firewall. **D**, VLAN segmentation does add administrative overhead, but this is weighed against a documented, measurable reduction in lateral movement risk. The NIST Cybersecurity Framework and SP 800-53 (SC-7 Boundary Protection) identify internal segmentation as a substantive control, not a cosmetic one.*

8. A network architect proposes layering zero-trust principles on top of existing VLAN boundaries. A colleague argues that VLANs already provide adequate segmentation and zero trust adds unnecessary complexity. Which response most accurately evaluates this disagreement?

○ A. The colleague is correct: VLANs fully enforce zero trust because devices in different VLANs cannot communicate by default.

● B. The architect is correct because VLANs create coarse-grained perimeters, zero trust adds per-session identity and authorization verification that VLAN boundaries alone cannot enforce.

○ C. The architect is correct because zero trust eliminates the need for VLANs, making network segmentation redundant.

○ D. Both are incorrect: zero trust is an application-layer security model and does not interact with network-layer segmentation.

Correct answer: B

Why: VLANs create macro-segmentation: broad zones in which all devices can freely communicate with each other without further challenge. Zero trust applies the principle of "never trust, always verify" at a finer level, every individual connection request, even between devices in the same VLAN, must be authenticated and authorized against a policy engine before it is permitted. NIST SP 800-207 (Zero Trust Architecture) explicitly frames VLANs as a legacy segmentation mechanism that zero trust supplements, not replaces. The two controls are complementary: VLANs establish network-level zones; zero trust enforces identity-based micro-segmentation within and across those zones.

*Why the others are wrong: **A**, VLANs block inter-VLAN traffic by default, but all devices within a VLAN can communicate freely with each other. Zero trust requires per-session verification even within a zone, a property VLANs do not provide. **C**, Zero trust is a security model, not a physical or logical network topology replacement. Network segmentation (including VLANs) remains a relevant and recommended control within a zero trust architecture, NIST SP 800-207 describes how they coexist. **D**, Zero trust is a cross-layer model. While it encompasses application-layer authorization, it also directly addresses network-layer access decisions, device health verification, and micro-segmentation, all of which interact with network segmentation policy.*

9. A naval submarine network technician manages three LANs: one for weapons systems, one for official naval operations, and one for crew recreation. The technician must recommend segmentation controls proportional to each LAN's sensitivity level. Which design is most defensible from a security standpoint?

○ A. Merge the weapons and operations LANs into one high-security VLAN and isolate only the recreation LAN, reducing administrative complexity while consolidating the most sensitive systems.

○ B. Connect all three LANs to a central switch without routing restrictions but install an intrusion detection sensor on the weapons LAN only.

● C. Maintain three fully isolated segments: air-gap or hardware-firewall the weapons LAN from all other traffic, enforce strict access controls between operations and recreation, and treat recreation traffic as untrusted.

○ D. Allow the operations LAN to communicate freely with both the weapons and recreation LANs, since all crew members accessing those LANs hold appropriate clearances.

Correct answer: C

Why: The weapons LAN controls safety-critical, potentially lethal systems. Even a single unauthorized command issued from another network segment, whether from a compromised operations workstation or a crew member's recreation device, could have catastrophic consequences. An air gap or hardware-enforced firewall boundary provides the strongest isolation and cannot be bypassed by software misconfiguration. The recreation LAN must be treated as the lowest-trust segment because crew members use it for personal internet access, maximizing its exposure to malware and social engineering. Merging the weapons and operations LANs (Option A) removes a critical boundary between systems of different sensitivity. Option D conflates personnel clearance with technical access entitlement, a violation of the principle of least privilege that zero trust frameworks (NIST SP 800-207) and classical defense-in-depth models both reject.

*Why the others are wrong: **A**, Merging weapons and operations LANs eliminates the boundary between them. A compromise on an operations workstation, via phishing, unpatched software, or a malicious attachment, would then have direct network access to weapons systems, which is operationally and strategically unacceptable. **B**, Connecting all three LANs to a central switch without routing restrictions creates a flat network; no segmentation exists. IDS sensors detect attacks in progress but do not prevent cross-LAN lateral movement. This design provides detection without containment, the weakest possible architecture for this threat environment. **D**, Personnel clearance is an administrative and physical access control. It does not establish technical network boundaries. An authorized person's device can still be*

compromised; technical segmentation enforces access regardless of who (or what malware) is using the device.